UNIVERSITI TEKNOLOGI MALAYSIA

LAB REFLECTION 1 Team 1 (7.5%)

SEMESTER II 2021/2022

<<LIST ALL TEAM MEMBERS' NAME WHO HAVE PARTICIPATED IN THE LAB ONLY>>

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CODE OF SUBJECT: SECJ3403

NAME OF SUBJECT: SPECIAL TOPIC IN SOFTWARE ENGINEERING

OBJECTIVES:

1. Identify the important concepts in cloud computing.
2. Do reflection on practical lab implementation based on the concepts identified.

INSTRUCTION:

1. Create reflection on practical lab implementations for the following labs:
    i. Compute
    ii. Storage

Reflection on practical lab implementations (based on Gibbs' Reflective Model)

The following templates should be used to reflect on practical lab implementation. You have to use the same templates for all lab reflections.

Reflection on Lab 3 - Introduction to Amazon EC2	
Description	Describe in brief what you did in the lab in your own words. ● In this lab, we were introduced to accessing AWS management console to create an Amazon EC2 instance. ● A step-by-step guide is provided to explain in more detail what this lab tries to accomplish. ● I followed the instructions on the lab 3 introduction to Amazon EC2 modules and found it to be easy to follow. ● A successful EC2 setup at the end of the lab session.
Evaluation	Describe the positive or negative experience during the lab ● Hands-on experience by individuals is importantly gained by the student itself. ● A good introduction starts for a student that does not have any experience in managing cloud architecture. ● A complete guide helped students to be at ease in completing the lab. ● The positive experience of trying the lab was discovering features provided by AWS that I can replicate for my own application. The other positive experience was learning how to use the console and familiarizing myself with AWS UI. ● The negative experience was searching for buttons and tabs that were described in the instructions. Since I am new with the UI, I have to re-read the instructions and re-try several times just to get the same result as the one in the instructions.

Analysis	i. Which practical implementation in the lab can be related to the theory in the lecture? <u>EC2</u> ● The lecturer did explain everything in detail about all the things that are available in the console for example the Availability Zone. By following the instructions, we understand fully how the console works. ● The Lecturer explains in detail how EC2 enables the user to use a virtual server as many as they like ● All contents in the learning lab related to the theory given during lecture-discussion and helped us to set up EC2 in Amazon. <u>Storage</u> ● The lecturer also explains in detail the storage types in AWS. We know which storage to choose and why we chose them by following the instructions. ● We know how AWS storage store and retrieve the data based on what the lecturer provide ● All contents in the learning lab related to the theory given during lecture-discussion and helped us to set up Storage in Amazon. ii. What other concepts in your opinion should be included in the lab implementation? <u>EC2</u> ● A serverless concept such as FaaS should be included because we can directly develop an app and deploy functions e.g. business logic without maintaining a server. ● From my point of view, if we are talking about EC2, it is also important to show a lab about Remote RDP set up so we can fully utilize the benefit of EC2 instance.

	<u>Storage</u> ● Concept of data storage should be included so that we can know the difference between file storage, block storage, and object storage. ● Tutorial labs of different storage types are quite important to be shown so it can help us as students to fully understand and be able to decide on which scenario or situation that specific storage type meets the criteria to be chosen. ● Understanding the different types of storage is also important so we know how fast, safe and efficient each storage is when doing its task
Conclusion	Focus on what you have learned. Are there any skills you developed because of the experience? If so, how would you apply them in future experiences or situations? ● By learning in this lab, students are able to develop new knowledge in EC2 and Storage not just in theory but also in practical use. Help to boost productivity and creativity in the current work environment in the IT field. ● From this experience, students are able to develop a virtual server that can help reduces the physical server in a company on their premises ● We are able to familiarize ourselves with the AWS services. If we ever have the chance to use cloud services, we might as well use AWS because we can leverage our experience from this course to help accelerate our future projects.
Action Plan	What specific implementation in this lab should be included in your proposed team project? Project: Mersing tourism

	<u>EC2</u> ● At the moment, there is no implementation in AWS microservice for the work environment. Lab practices are needed to maintain the skill developed during this lab. If we are able to extend the use of this AWS Cloud Foundation, we may be able to fully understand the AWS ecosystem and help us move our project to AWS. ● If we can use AWS EC2 for free we might use it to create instances and install the webserver of our choice which is Xampp web server. <u>Storage</u> ● Our current situation is we use monolithic architecture where we store all storage data in a single same server as our web server. If we are able to prolong lab practice to fully understand the AWS ecosystem, we are able to decide to move our data to AWS storage. ● If we can use AWS S3(simple storage) we can use it to store a large amount of data. ● If we can use AWS EBS (elastic block store) for free we might use it to host our PHP files and at the same time, we can learn how block storage works.

Rubric for reflection

Criteria	**Excellent**	**Good**	**Fair**	**Unsatisfied**
Evaluation **(5 marks)**	Clearly explains positive experience during lab that went particularly well and negative aspects of the lab experience that did not go	Somewhat explains positive experience during lab that went particularly well and negative aspects of the lab experience that did not go well or did not work.	Little explanation on positive experience during lab that went particularly well and negative aspects of the lab experience that did not go well or did not work.	No clear explanations on positive experience during lab that went particularly well and negative aspects of the lab experience that did not go well or did not work.

	well or did not work. **(4-5 marks)**	**(3-3.9 marks)**	**(2-2.9 marks)**	**(0-1.9 marks)**
Analysis **(10 marks)**	Analysis shows strong understanding between the concept and the implementation **(8-10 marks)**	Analysis shows mostly strong understanding between the concept and the implementation. **(5 -7.9 marks)**	Analysis shows little evidence of understanding between the concept and the implementation **(3 -4.9 marks)**	Analysis shows no evidence of understanding between the concept and the implementation **(0 -2.9 marks)**
Conclusion **(5 marks)**	Conclusion shows strong evidence of synthesis of ideas presented and insights gained throughout the entire course. **(4-5 marks)**	Conclusion mostly shows strong evidence of synthesis of ideas presented and insights gained throughout the entire course. **(3-3.9 marks)**	Conclusion shows little evidence of synthesis of ideas presented and insights gained throughout the entire course. **(2-2.9 marks)**	Conclusion shows no evidence of synthesis of ideas presented and insights gained throughout the entire course. **(0-1.9 marks)**
Action Plan **(5 marks)**	Thoroughly explains importance of the action and future implementation in the group project. **(4-5 marks)**	Somewhat explains importance of the action and future implementation in the group project. **(3-3.9 marks)**	Unclearly explains importance of the action and future implementation in the group project. **(2-2.9 marks)**	No explanation of the importance of the action and future implementation in the group project. **(0-1.9 marks)**

Writing of the Reflection **(5 marks)**	Writing is clear, concise, and well organized with excellent sentence/paragraph construction. Thoughts are expressed in a coherent and logical manner. There are no more than three spelling, grammar, or syntax errors per page of writing. The documentation is in systematic and formatted manner. **(4-5 marks)**	Writing is mostly clear, concise, and well organized with good sentence/paragraph construction. Thoughts are expressed in a coherent and logical manner. There are no more than five spelling, grammar, or syntax errors per page of writing. The documentation is in somewhat systematic and formatted manner. **(3-3.9 marks)**	Writing is unclear and/or disorganized. Thoughts are not expressed in a logical manner. There are more than five spelling, grammar, or syntax errors per page of writing. The documentation is a little systematic and formatted manner. **(2-2.9 marks)**	Writing is unclear and disorganized. Thought's ramble and make little sense. There are numerous spelling, grammar, or syntax errors throughout the response. The documentation has no systematic and formatted manner. **(0-1.9 marks)**